

Lesson 3:

Using R to process data

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Outlines

- Coding by reviewing the website: <https://chenziqu0506-ai.github.io/ziqichenRsimpletutorial/lesson3.html>
- Materials also could be downloaded via the above link:

6. 分析 diamonds 数据集

通过 diamonds 数据集实践数据统计技能。数据集包含钻石的克拉数、切工、价格等信息，可用于练习选取、更改、统计、分组和合并操作。

资源：

- [下载 diamonds 数据集 \(CSV\)](#)
- [下载数据统计 R 代码](#)

Left_join()

- 假设我们有两个不同的数据集

	name	mean_score
	<chr>	<dbl>
1	Alice	73.5
2	Bob	78
3	Carol	90.5

成绩

	name	age
	<chr>	<dbl>
1	Alice	12
2	Bob	13
3	Dave	14

年龄

Left_join()

- 假设我们有两个不同的数据集

	name	mean_score
	<chr>	<dbl>
1	Alice	73.5
2	Bob	78
3	Carol	90.5

成绩

	name	age
	<chr>	<dbl>
1	Alice	12
2	Bob	13
3	Dave	14

年龄

	name	mean_score	age
	<chr>	<dbl>	<dbl>
1	Alice	73.5	12
2	Bob	78	13
3	Carol	90.5	NA

Left_join(成绩, 年龄, by = "name")

Right_join()

- 假设我们有两个不同的数据集

```
name  mean_score
<chr>    <dbl>
1 Alice    73.5
2 Bob      78
3 Carol    90.5
```

成绩

```
name  age
<chr> <dbl>
1 Alice    12
2 Bob      13
3 Dave     14
```

年龄

```
# A tibble: 3 x 3
  name  mean_score  age
<chr>    <dbl> <dbl>
1 Alice    73.5    12
2 Bob      78      13
3 Dave     NA      14
```

Right_join(成绩, 年龄, by = "name")

Full_join()

- 假设我们有两个不同的数据集

	name	mean_score
	<chr>	<dbl>
1	Alice	73.5
2	Bob	78
3	Carol	90.5

成绩

	name	age
	<chr>	<dbl>
1	Alice	12
2	Bob	13
3	Dave	14

年龄

	name	mean_score	age
	<chr>	<dbl>	<dbl>
1	Alice	73.5	12
2	Bob	78	13
3	Carol	90.5	NA
4	Dave	NA	14

Full_join(成绩, 年龄, by = "name")

Inner_join()

- 假设我们有两个不同的数据集

```
name  mean_score
<chr>    <dbl>
1 Alice    73.5
2 Bob      78
3 Carol    90.5
```

成绩

```
name  age
<chr> <dbl>
1 Alice    12
2 Bob      13
3 Dave     14
```

年龄

```
# A tibble: 2 × 3
  name  mean_score  age
<chr>    <dbl> <dbl>
1 Alice    73.5    12
2 Bob      78     13
```

Inner_join(成绩, 年龄, by = "name")

筛选联结 `semi_join()`; `anti_join()`

- 假设我们有两个不同的数据集

	name	mean_score
	<chr>	<dbl>
1	Alice	73.5
2	Bob	78
3	Carol	90.5

成绩

	name	age
	<chr>	<dbl>
1	Alice	12
2	Bob	13
3	Dave	14

年龄 成绩 `%>% semi_join(年龄, by = "name")`

	name	mean_score
	<chr>	<dbl>
1	Alice	73.5
2	Bob	78

筛选联结 `semi_join()`; `anti_join()`

- 假设我们有两个不同的数据集

	name	mean_score
	<chr>	<dbl>
1	Alice	73.5
2	Bob	78
3	Carol	90.5

成绩

	name	age
	<chr>	<dbl>
1	Alice	12
2	Bob	13
3	Dave	14

年龄

	name	mean_score
	<chr>	<dbl>
1	Carol	90.5

成绩 %>% `anti_join(年龄, by = "name")`

Let's do some exercises!

Thank you!

For more information:

- <https://chenziqu0506-ai.github.io/ziqichenRsimpletutorial/index.html>
- <https://github.com/chenziqu0506-AI/ziqichenRsimpletutorial/tree/main>